HUM64-3

Sustainable Futures

Reasoning honestly under planetary constraints

Neither naïve optimism nor reflexive catastrophism: quantitative reasoning under uncertainty.

Albert School — 12 hours

Contents

1 Planetary boundaries

The planetary-boundaries framework (Rockström et al.) defines a set of biophysical processes for which a quantitative control variable and a proposed "safe operating space" can be stated. Crossing a boundary is held to raise the risk of large-scale, possibly abrupt, environmental change.

> **Definition 1** A **planetary boundary** is a threshold value of a control variable for an Earth-system process, below which the process is judged to remain in a Holocene-like state and above which the risk of destabilisation rises.

The nine processes are: climate change, biosphere integrity (genetic and functional), land-system change, freshwater change, biogeochemical flows (nitrogen and phosphorus), ocean acidification, atmospheric aerosol loading, stratospheric ozone depletion, and novel entities.

Critical treatment is required: the boundaries differ in how defensibly they can be measured.

- **Well-measured and defensible.** Climate change (control variables: atmospheric CO_2 concentration and radiative forcing), stratospheric ozone depletion, and ocean acidification rest on a single global control variable with a clear physical basis.
- **Contested.** Biosphere integrity (no agreed functional metric), the aggregate nitrogen and phosphorus boundaries (strongly regional, not global), freshwater change, and novel entities are harder to quantify; the boundary value is a judgement that depends on the chosen variable, so the precise number is open to dispute.

Common pitfall A crossed boundary is a statement about elevated risk, not a dated prediction of collapse. The framework's quantitative claims are strongest where a single global control variable exists and weakest where the control variable is regional or undefined.

2 Climate science at decision resolution

This chapter reads the IPCC (GIEC) Sixth Assessment Report (AR6) at the resolution needed for decisions, not for general awareness.

2.1 Carbon budgets

Definition 2 A **carbon budget** is the cumulative quantity of CO_2 that may still be emitted while keeping global warming below a stated temperature limit with a stated probability.

The near-linear relationship between cumulative CO_2 emissions and peak warming is what makes a budget meaningful: each tonne emitted draws down a finite remaining stock. AR6 (assessed from the beginning of 2020) gives, approximately:

- **1.5°C**, 50% probability: about 500 $GtCO_2$ remaining.
- **2°C**, 67% probability: about 1150 $GtCO_2$ remaining.

These numbers fall by roughly the current annual emissions (about 40 $GtCO_2$) each year, and they shift with the chosen probability and with non-CO_2 forcing assumptions.

2.2 Warming scenarios and uncertainty

AR6 organises projections around the Shared Socioeconomic Pathways (SSPs), from very low (SSP1-1.9) to very high (SSP5-8.5) emissions. For each scenario the report states a **best estimate** and a **very likely range** of warming. The range has two sources: scenario uncertainty (which emissions path is followed) and physical uncertainty (the climate's sensitivity to a given forcing, summarised by the equilibrium climate sensitivity, assessed best estimate 3°C, likely range 2.5–4°C).

Common pitfall A decision that is robust only at the best estimate but fails across the very likely range is not robust. The range is the decision-relevant object, not the central number.

2.3 Tipping points

Tipping elements are components of the climate system that may pass a threshold beyond which change becomes self-sustaining. The evidence is uneven, and established cases must be separated from speculative ones.

- **Permafrost.** Thaw releasing CO_2 and CH_4 is established as a positive feedback; its magnitude and timing are uncertain.
- **Greenland ice sheet.** Loss is observed and a long-run threshold for irreversible decline is plausible; full loss unfolds over centuries to millennia.
- **AMOC (Atlantic Meridional Overturning Circulation).** Weakening is observed; a full collapse this century is assessed as low-probability but high-impact, and the threshold is not well constrained — the most speculative of the three.

3 Reading a climate-scenario output

A climate or energy scenario is a conditional projection: **if** a set of assumptions holds, **then** these outcomes follow. Reading one means recovering the assumptions, not consuming the headline.

The IEA **Net Zero by 2050** (NZE) scenario is a normative pathway: it fixes the outcome (net-zero energy-sector CO_2 by 2050, consistent with 1.5°C) and works backwards to the actions required.

To read such an output, identify three things.

1. **Structural assumptions.** The drivers the scenario imposes — e.g. technology cost trajectories, the pace of electrification, demand reduction and energy efficiency, the availability of carbon capture and of bioenergy, and behavioural change. The result is a consequence of these inputs.
2. **Uncertainty ranges.** Where the scenario is sensitive to an assumption (e.g. the scale of carbon dioxide removal it relies on), and how a different assumption would move the path.
3. **Sector implications.** What the pathway requires of a named sector — for the NZE, for example, no new unabated coal plants and no new oil and gas fields approved for development beyond those already committed, on its stated assumptions.

Common pitfall A normative scenario (NZE) states what would be required to reach a target; it is not a forecast of what will happen. Confusing the two misreads the output.

4 Energy-transition economics

4.1 LCOE

Definition 3 The **levelised cost of electricity (LCOE)** is the constant per-unit price of electricity that sets the discounted revenue of a generating asset equal to its discounted lifetime cost:

$$\mathrm{LCOE} = \frac{\sum_{t=0}^{T} \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=0}^{T} \frac{E_t}{(1+r)^t}},$$

where I_t is investment, M_t operations and maintenance, F_t fuel cost, E_t electricity generated in year t, r the discount rate, and T the asset lifetime.

LCOE collapses an asset's whole-life economics into one number, which makes it comparable across technologies but also makes its omissions matter.

What LCOE **captures**: capital cost, operating and fuel cost, lifetime output, and the cost of capital through r.

What LCOE **misses**:

- **Intermittency.** A unit from a dispatchable plant and a unit from a variable source (wind, solar) are not equivalent; LCOE treats them as identical.
- **Grid and integration costs.** Transmission, balancing, and curtailment are not in the per-asset figure.
- **Storage.** The cost of firming variable output is external to LCOE.

Definition 4 **Integration cost** is the system-level cost of accommodating a generation source — grid reinforcement, balancing reserves, and storage — that LCOE omits. It rises with the share of variable renewables on the system.

4.2 The renewable cost learning curve

A **learning curve** states that unit cost falls by a roughly constant percentage for each doubling of cumulative installed capacity (the learning rate). Solar PV and batteries have followed steep learning curves, and their realised cost falls repeatedly came in below the projections of historical models, which assumed slower or no learning. The decision implication: models that extrapolate past costs linearly have systematically under-predicted renewable deployment, and allocation decisions made on them under-weighted renewables.

Common pitfall Learning-curve gains lower generation LCOE but do not remove integration cost; at high penetration, integration cost can rise even as LCOE falls.

5 Corporate sustainability strategy

5.1 Net Zero as constraint vs. signal

A corporate **Net Zero** commitment is either a binding constraint on the firm's decisions or a marketing signal. The distinction is whether the commitment carries a dated mechanism and near-term milestones, or only a distant headline year. The credibility critique is developed in the final chapter.

5.2 Scopes 1, 2, 3

The GHG Protocol partitions a firm's emissions into three scopes.

Definition 5
- **Scope 1** — direct emissions from sources the firm owns or controls.
- **Scope 2** — indirect emissions from purchased energy (electricity, heat).
- **Scope 3** — all other indirect emissions in the value chain, upstream (purchased goods, transport) and downstream (use and end-of-life of sold products).

Scope 3 is the central and hardest challenge: for most firms it is the largest share of the footprint, yet it lies outside the firm's direct control, depends on supplier and customer data the firm does not own, and is prone to double counting across firms. A Net Zero claim that omits Scope 3 typically omits most of the emissions.

6 Climate finance

6.1 Physical vs. transition risk

Definition 6
- **Physical risk** — financial loss from the physical effects of climate change: acute (storms, floods, wildfires) and chronic (sea-level rise, heat, drought).

- **Transition risk** — financial loss from the move to a low-carbon economy: policy and legal change, technology substitution, market shifts, and reputation, including stranded assets.

The two can move in opposite directions: faster transition lowers long-run physical risk but raises near-term transition risk. Both are integrated into valuation in the next chapter on S6 integration.

6.2 ESG rating architecture and its critiques

ESG ratings score firms on Environmental, Social, and Governance dimensions. Legitimate critiques:

- **Inter-agency divergence.** Ratings of the same firm by different agencies correlate weakly, because agencies differ in what they measure, how they weight it, and how they fill missing data — unlike credit ratings, which agree closely.
- **Greenwashing.** A rating can reward disclosure and policy language rather than realised outcomes, so a firm can score well by reporting well.
- **The 'S' is hard to measure.** Social factors (labour, community, human rights) lack agreed quantitative metrics, so the 'S' is the least comparable pillar.

6.3 Evidence on ESG and impact

- **ESG strategy performance.** The empirical evidence that ESG-screened portfolios systematically out- or under-perform is mixed and sensitive to how the strategy is constructed; there is no robust general result.
- **Impact-investing additionality.** The relevant question is **additionality** — whether the investment caused an impact that would not otherwise have occurred. Buying a share on the secondary market need not change a firm's behaviour, so reported impact is often not additional.

7 Intergenerational reasoning

7.1 The discount rate as an ethical choice

Climate costs and benefits fall across generations, so they must be discounted to be compared. The discount rate is not purely technical; its main component is an ethical choice about the weight given to future people.

The Ramsey rule expresses the consumption discount rate as

$$r = \delta + \eta g,$$

where δ is the pure rate of time preference (how much future welfare is discounted simply for being future), η the elasticity of marginal utility of consumption, and g the growth rate of consumption per capita.

Definition 7

- **Stern (2006)** sets δ close to zero on ethical grounds — future generations' welfare counts almost equally — giving a low discount rate and a case for large immediate mitigation.
- **Nordhaus** calibrates δ to observed market returns, giving a higher discount rate and a case for more gradual mitigation.

The two reach opposite policy conclusions largely from the choice of δ, not from different climate science. The choice should therefore be made explicitly, as an ethical position.

7.2 Climate justice

Distributional questions, to be treated with quantitative honesty:

- **Historical emissions.** Cumulative emissions, not current flows, drive warming; the regions that emitted most historically (and per capita) differ from those most exposed to impacts.
- **Incidence of transition cost.** Who bears the cost of decarbonisation — across countries and across income groups within a country — is a separate question from who caused the problem.
- **Obligations.** What follows from historical responsibility and from differential capacity is an explicit normative claim, to be argued rather than assumed.

8 S6 integration

Sustainability enters strategic and financial decisions as a **binding input**, not as separate compliance reporting. Concretely, a sustainability-constraint analysis for a real strategic decision integrates:

- **Climate data** — the relevant carbon budget, scenario, or physical exposure.
- **Economic mechanisms** — LCOE and integration cost, learning curves, the cost of transition.
- **Financial implications** — physical and transition risk carried into a valuation (drawing on Valuation & Investment Analysis, BUS43-3).
- **Interactions** with Strategic Allocation (BUS63-4), Corporate Finance Leadership (BUS63-1), and Public Policy (HUM64-2): the same constraint reappears as an allocation limit, a cost of capital and capital-structure input, and a policy parameter.

The deliverable is an analysis that can be defended: it states its climate assumptions, traces them through the economic and financial mechanisms, and reaches a decision the constraints support.

9 Net Zero credibility critique

A Net Zero commitment is assessed by separating binding elements from decorative ones.

Definition 8

- **Binding** — targets validated against an external standard (e.g. SBTi — Science Based Targets initiative) with dated, mechanism-level near-term milestones covering Scopes 1, 2, and the material part of Scope 3.
- **Decorative** — "carbon neutrality" claimed through purchased offsets without a corresponding reduction in the firm's own emissions, or a distant target year with no near-term mechanism.

The test:

1. Is there a near-term (e.g. 2030) target, not only a distant year?
2. Is it externally validated (SBTi or equivalent)?
3. Does it cover Scope 3 where Scope 3 is material?
4. Does it rest on **reduction**, with offsets used only for genuinely residual emissions, rather than substituting offsets for reduction?

A commitment failing these tests is a marketing signal, not a constraint.